

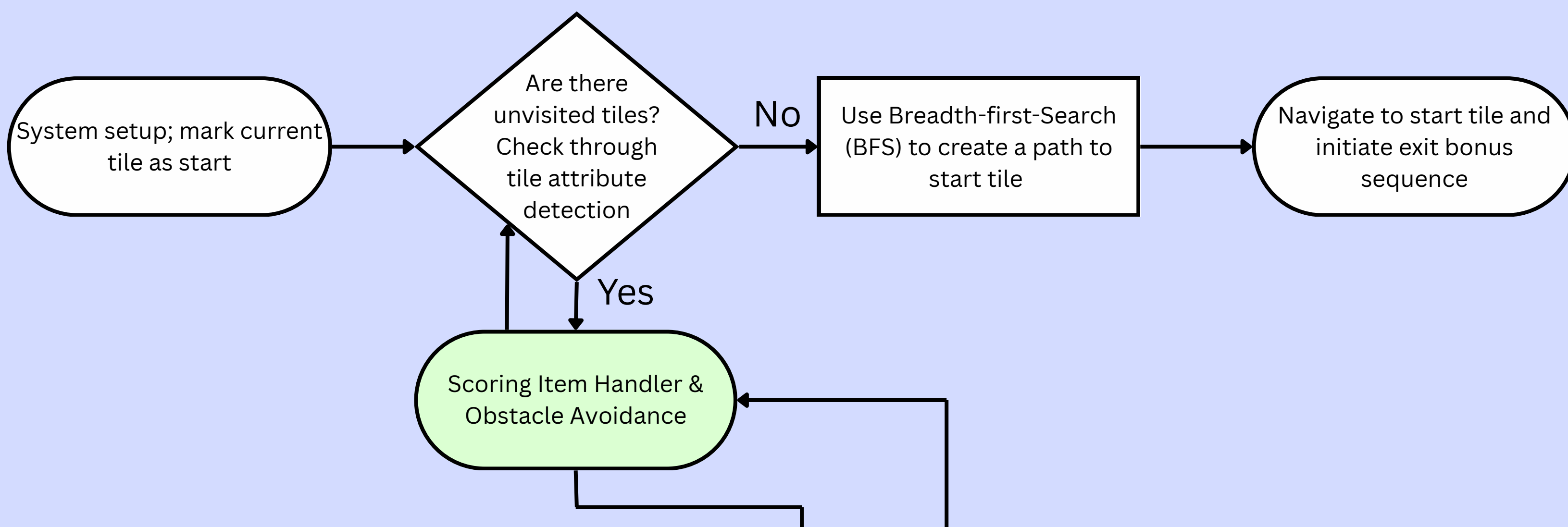


United States of America

Pi-rates - RCJ Rescue Maze 2026



Software Architecture



Localization Error

```
<tolerance_dist> = 3
tof1 = ( wall on right ? back_tof : front_tof )
tof2 = ( wall on left ? front_tof : back_tof )

while | difference = the front-tof, back-tof | > tolerance_dist
  turn (tof1 > tof2) until the difference < tolerance_dist
```

Issue: IMU was unreliable for navigation and was found to drift about 2°/minute - un

Solution - Self-alignment for error correction:

- The robot aligns to straight walls by comparing readings from two Time-of-Flight sensors.
- Adjusts its angle until both sensors report equal distances, indicating parallel alignment.
- Front wall detection is used to maintain proper tile spacing and prevent collisions.

Colored tile detection

Obstacle Avoidance

Localization Error

Ramp Detection via IMU

Victim Detection
(Computer Vision)

Obstacle Avoidance

Robot

Obstacle

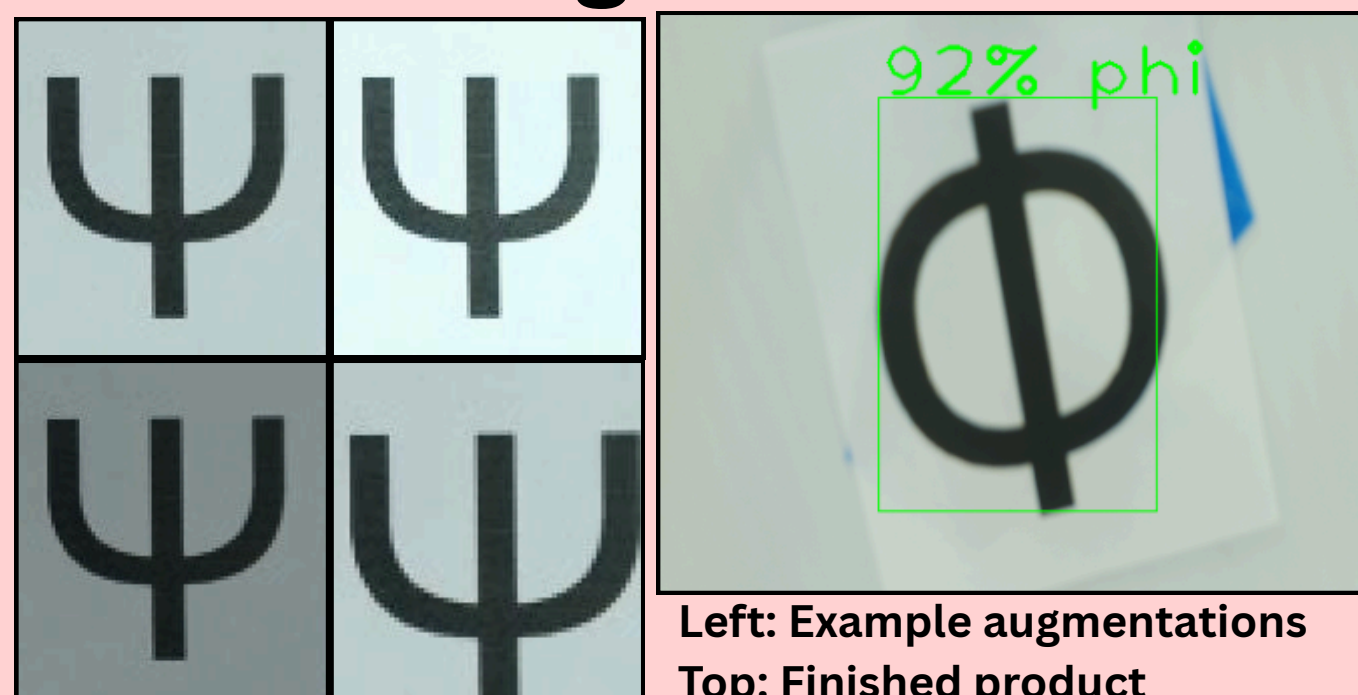
Lines = ToF infrared beam
Right sensor < left and center
Also less than 1 tile → obstacle on right of next tile

Issue: Need premature and accurate info on where obstacle is in order to mark it in file.

Solution - Comparative localization:

- Robot compares distance in all 3 ToF sensors
 - All 3 report same dist → wall
 - Obstacle can be located based on which sensor returns lowest distance

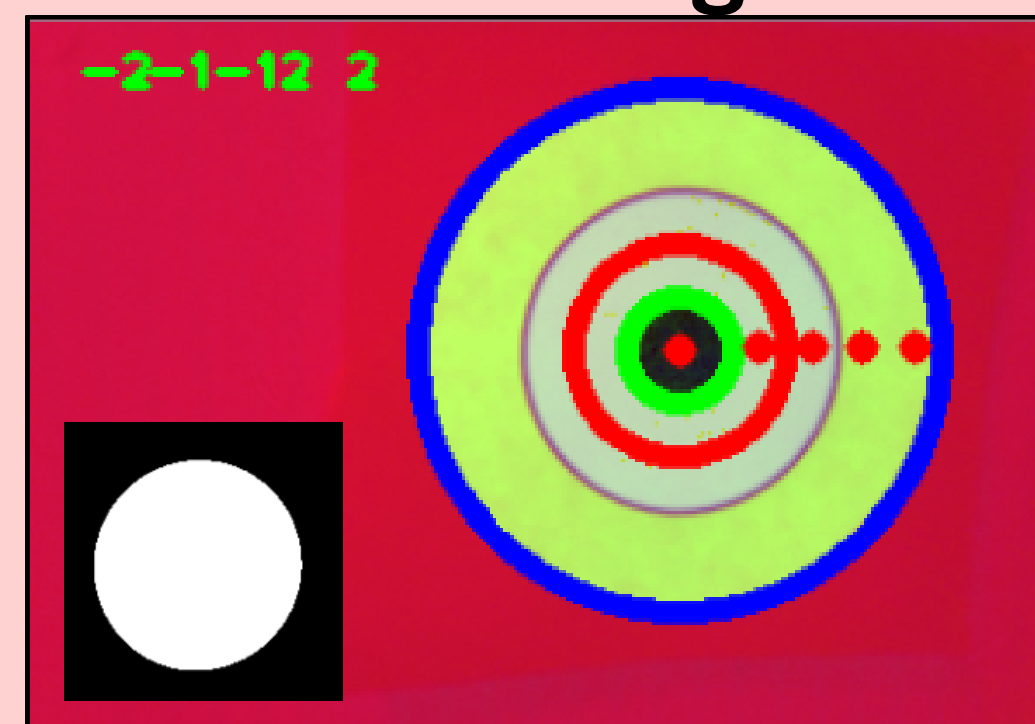
Letter Recognition (YOLO)



Left: Example augmentations
Top: Finished product

The You Only Look Once (YOLO) framework creates predictive bounding boxes based on user-made training images. To improve its accuracy, we added **augmentations** to the training data, such as brightness changes and zoom. **Zooms** simulate the robot being closer/further from the letter, and **brightness** simulates different light levels and shadows. These augmentations help the trained YOLO model become more robust.

Cognitive Targets



Ring values:
-2, -1, -1, 2, 2
Black, Red, Red, Blue, Blue

Sum = 0 (Stable)
Blink LED but do not drop kits

We used **binary inverse thresholding** and HSV values to recognize the cognitive targets and sum up their rings. Binary inverse thresholding transforms the target's pixels to white, allowing a **contour** of the circle to be made. Then, the center and radius obtained from those operations are used to extract a pixel from each ring for its value.

Hardware



Alice Ma (Captain)
Software
Developed breath-first search, accumulated error, and obstacle avoidance algorithms



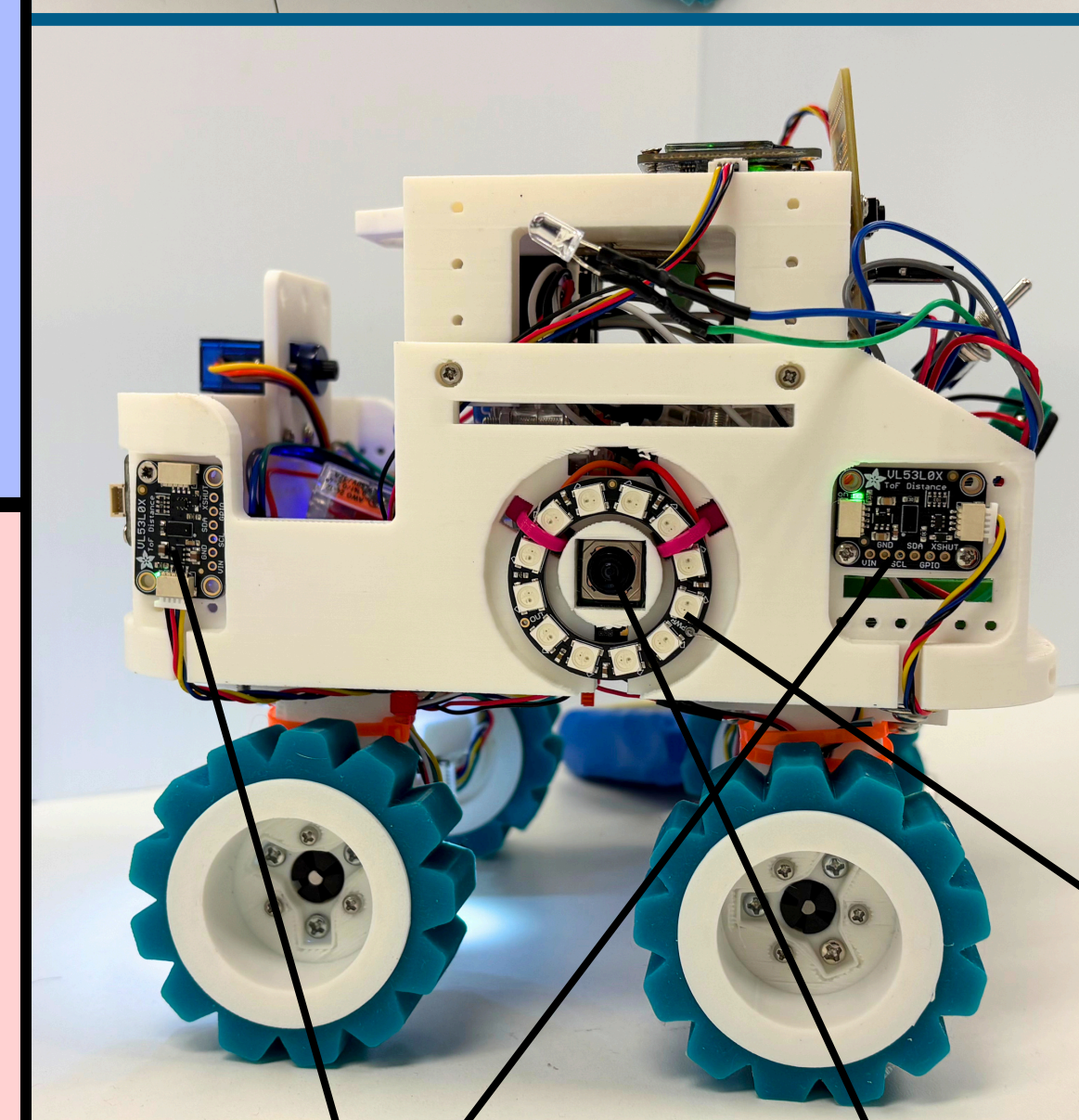
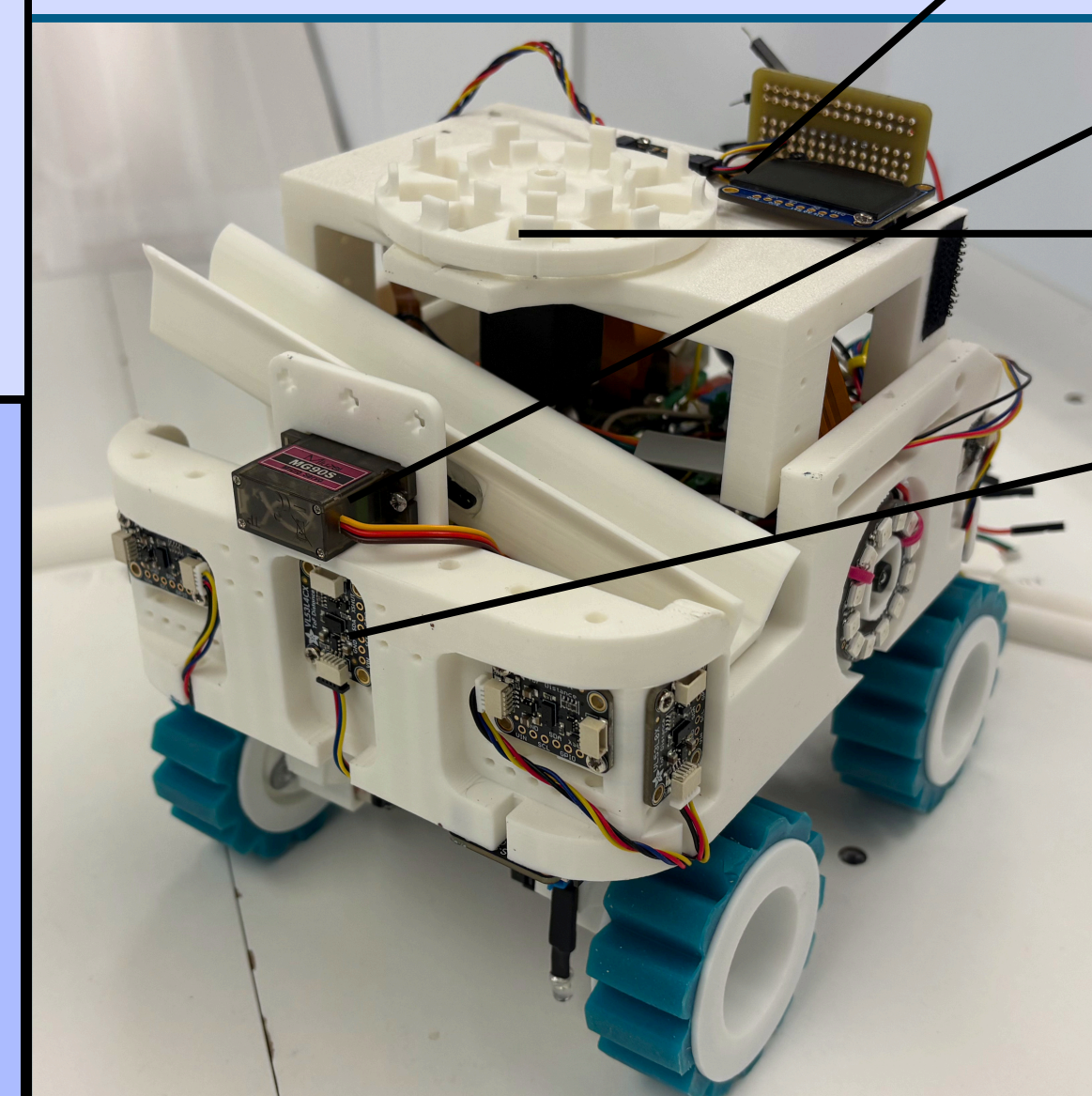
Showrya Verma
Software
Developed cognitive target and letter victim identification algorithms



Romit Gurao
Hardware
Designed robot chassis, designed dropper assembly, picked out parts & built robot



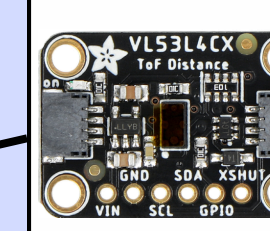
Rookie team
Won 1st competition as a team at RoboCup Junior USA 2026



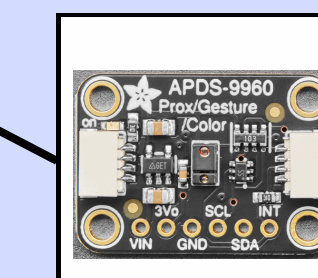
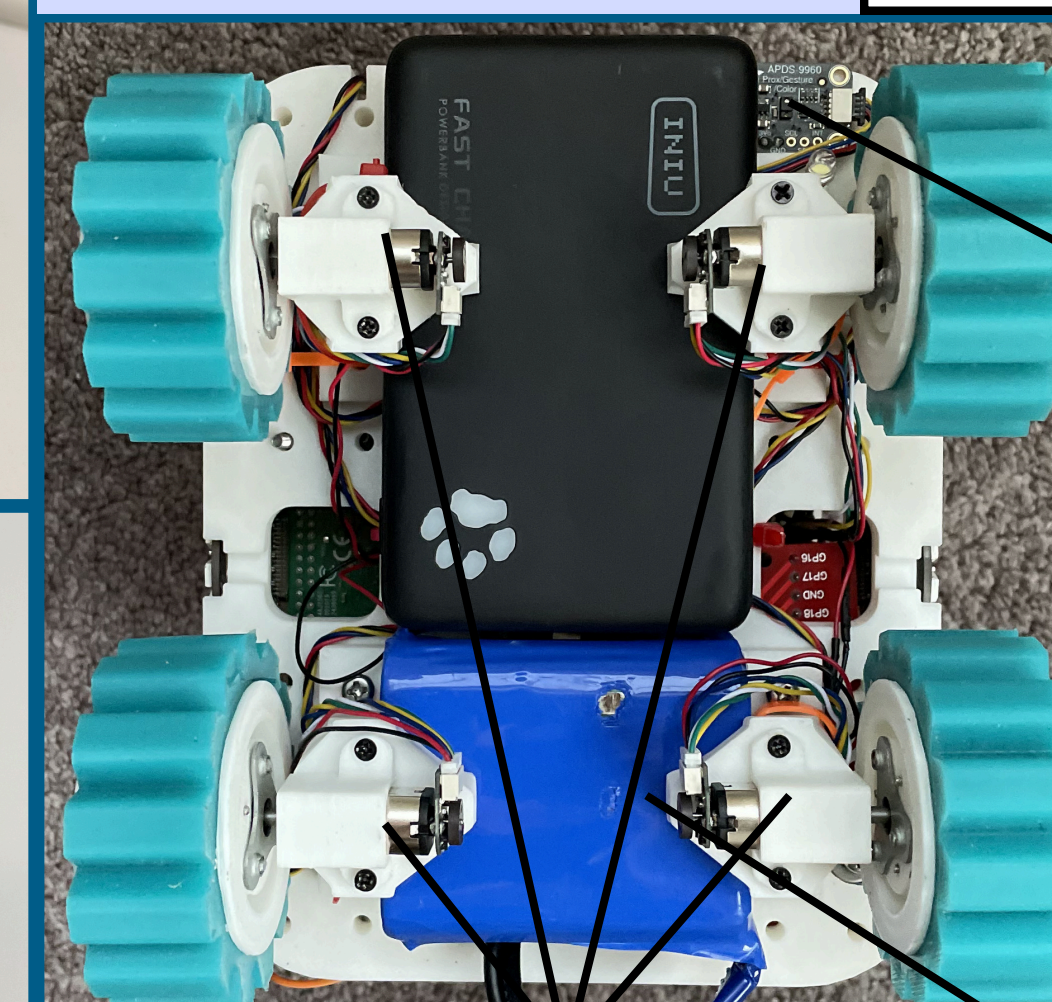
BNO055 IMU - Used for ramp detection & heading tracking



MG90D Servo Motor - Tilts rescue kit dispenser ramp



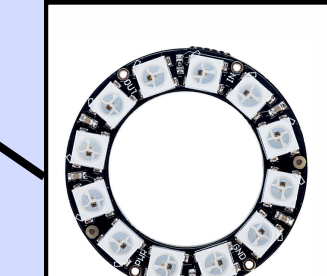
VL53L4CX Wide Angle ToF - Used to detect obstacles



APDS9960 Color Sensor - used in colored tile detection



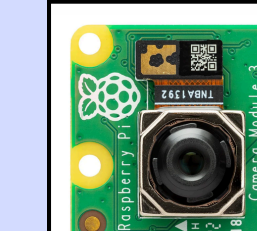
Pololu motors & encoders - Used for exact movement



Neopixel Light Ring - provides consistent lighting for cameras

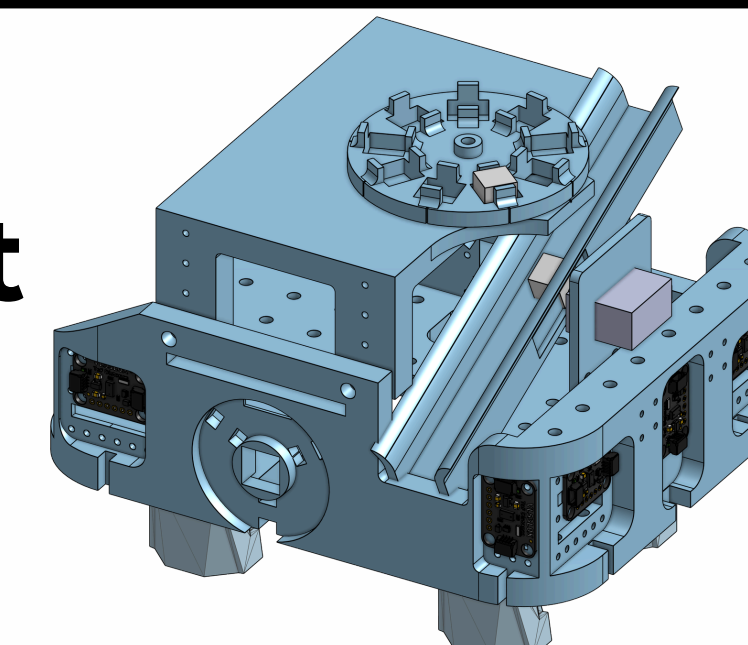


VL53L0X (x7) Time-Of-Flight-Wall / obstacle detection



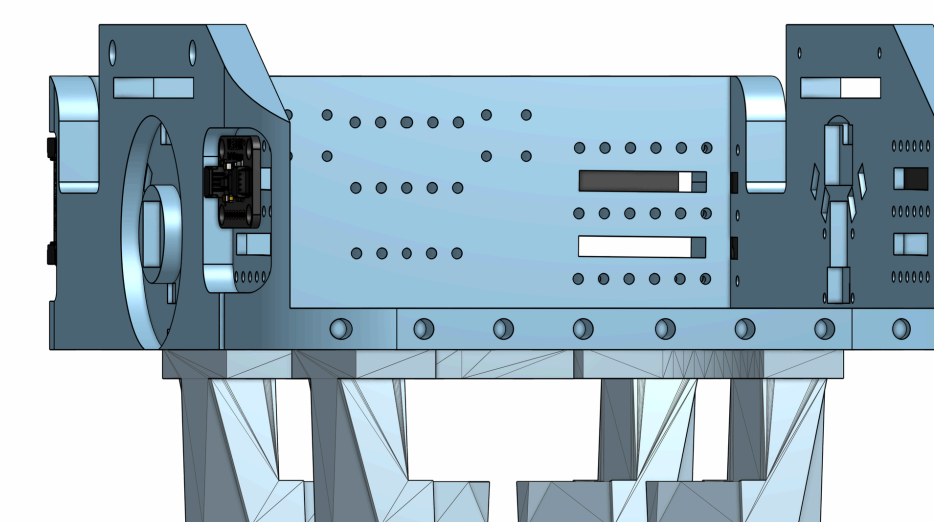
Raspberry Pi Camera (x2) - used for victim recognition

Rescue Kit Dropper



Rescue Kit Dropper Mechanism: Our robot utilizes a rotary rescue kit holder that drops kits through a hole when turned by a stepper motor. We then use a servo motor to tilt a ramp, after which the rescue kit slides down to the appropriate side. This mechanism was tested extensively for consistent and well-placed dropping.

Battery Mount



Battery Mount: Our robot's main battery is mounted on its underside, held in place by shelves on our motor mounts. This placement of the battery makes the robot's center of gravity lower, improving its capability to traverse ramps and stairs without flipping. It also makes the robot more compact.